

HEART DISEASE RISK PREDICTION

FEATURE GUIDE

Input fields • Codes • Units •
Meaning

Purpose	A quick-reference guide for entering the 11 patient features used by the trained machine-learning model.
Important	Enter values from reliable medical records or measurements. This application is for educational/predictive use and is not a medical diagnosis.

01 Quick Reference

#	Feature	What to enter	Code / Unit
1	Age	Completed age	Years
2	Sex	Male / Female	M / F
3	ChestPainType	Chest-pain category	ATA / NAP / ASY / TA
4	RestingBP	Resting systolic BP	mmHg
5	Cholesterol	Serum cholesterol	mg/dL
6	FastingBS	Fasting blood sugar	0 / 1
7	RestingECG	Resting ECG result	Normal / ST / LVH
8	MaxHR	Maximum heart rate	bpm
9	ExerciseAngina	Exercise-induced angina	N / Y
10	Oldpeak	ST depression value	Decimal
11	ST_Slope	ST segment slope	Up / Flat / Down

02 Patient Features

01 • Age

Meaning	Patient's completed age.
Why used	Age is an important demographic feature in cardiovascular-risk patterns.
Enter	Whole number only. Example: 45 . App range: 1–120 years .

02 • Sex

Meaning	Biological sex category used by the dataset.
Codes	M = Male F = Female
Enter	Select the appropriate option shown in the app.

03 • ChestPainType

Meaning	Category describing the patient's chest-pain pattern.
Codes	ATA = Atypical Angina • NAP = Non-Anginal Pain
More codes	ASY = Asymptomatic • TA = Typical Angina
Enter	Choose the category that matches the available clinical record.

04 • RestingBP

Meaning	Resting systolic blood pressure.
Unit	mmHg
Enter	Use the recorded resting systolic value. App range: 50–250 mmHg .

05 • Cholesterol

Meaning	Serum cholesterol measurement used in the dataset.
Unit	mg/dL
Enter	Use the recorded laboratory value. App range: 0–700 mg/dL .

06 • FastingBS

Meaning	Fasting blood sugar category.
Codes	0 = ≤ 120 mg/dL 1 = > 120 mg/dL
Enter	Select Normal (0) or High (1) in the app.

03 ECG & Exercise Features

07 • RestingECG

Meaning	Electrocardiogram result while resting.
Codes	Normal = normal ECG
More codes	ST = ST-T wave abnormality • LVH = Left Ventricular Hypertrophy
Enter	Select the recorded ECG category.

08 • MaxHR

Meaning	Maximum heart rate achieved/recorded.
Unit	bpm = beats per minute
Enter	Use the recorded maximum heart rate. App range: 50–250 bpm .

09 • ExerciseAngina

Meaning	Whether angina was present during exercise.
Codes	N = No Y = Yes
Enter	Select the option matching the exercise test/record.

10 • Oldpeak

Meaning	ST-segment depression associated with exercise relative to rest.
Format	Numeric value. Decimals are valid.
Examples	0.0, 0.5, 1.2, 2.5
Enter	Use the recorded value. App range: -3.0 to 10.0 .

11 • ST_Slope

Meaning	Slope of the peak exercise ST segment.
Codes	Up = Upsloping • Flat = Flat • Down = Downsloping
Enter	Choose the category matching the recorded result.

04 How the Prediction Works

STEP 1	Patient Input	The app collects 11 features from the form.
STEP 2	Preprocessing	Numerical and categorical values are transformed using the saved preprocessing pipeline.
STEP 3	Random Forest	The trained Random Forest model evaluates the processed feature set.
STEP 4	Probability	The app calculates the model probability associated with the heart-disease class.
STEP 5	Risk Band	The probability is displayed using application-defined Low, Moderate, or High presentation b

05 Prediction Result Bands

Probability	Displayed Result	Meaning in the Application
< 30%	LOW RISK	Lower model-predicted probability
30% to < 60%	MODERATE RISK	Intermediate model-predicted probability
>= 60%	HIGH RISK	Higher model-predicted probability

Note: These Low / Moderate / High bands are presentation categories defined by this application. They are **not clinical diagnostic thresholds**.

06 Data Entry Checklist

01	Use the patient's actual recorded values.
02	Do not guess coded categories such as ChestPainType, RestingECG, or ST_Slope.
03	Use the correct units: mmHg, mg/dL, and bpm.
04	Enter Oldpeak as a numeric value, including decimals when required.
05	For binary fields, select the exact meaning shown in the app: 0/1 or N/Y.
06	Review every field before clicking the prediction button.

07 Model Performance

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	90.22%	89.62%	93.14%	91.35%	93.31%
SVM	89.67%	88.79%	93.14%	90.91%	94.94%
KNN	89.67%	91.09%	90.20%	90.64%	94.31%
Logistic Regression	88.59%	87.16%	93.14%	90.05%	92.99%

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	79.35%	81.37%	81.37%	81.37%	79.10%

Selected Model	Random Forest
Selection basis	Highest F1-score among the evaluated models: 91.35% .

08 Disclaimer

Educational / Project Use Only

This application uses a machine-learning model to estimate a probability based on the supplied input features. The prediction is not a medical diagnosis, treatment recommendation, or substitute for a qualified healthcare professional. Clinical decisions should be made using appropriate medical evaluation and professional guidance.

Developed by
Jevin Kanani

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